

20 Mini C++ OOP Programs for Practice

- 1. Create a Student class with name and GPA. Add a function isDeanList() that returns true if GPA $\geq$ 3.5.
- 2. Create a BankAccount class with deposit(), withdraw(), and getBalance().
- 3. Create a Rectangle class that calculates area and perimeter.
- 4. Create a Circle class that calculates area and circumference.
- 5. Create a Car class with make, model, year, and a method getAge().
- 6. Create a Movie class with title, director, and year. Add a function isClassic() if year < 2000.
- 7. Create a Temperature class that converts between Celsius and Fahrenheit.
- 8. Create a Counter class that increments, decrements, and resets.
- 9. Create a Book class with title, author, and pages. Add a function isLongBook() if pages > 300.
- 10. Create a Point2D class with x and y and a function to calculate distance from origin.
- 11. Create a Product class with price and quantity. Add a function totalCost().
- 12. Create a Timer class that stores seconds and converts to minutes and hours.
- 13. Create a Person class with name and age. Add a function canVote().
- 14. Create a Triangle class that determines if it is equilateral, isosceles, or scalene.
- 15. Create a Password class that checks if a password meets strength requirements.
- 16. Create a Dice class that simulates rolling a die (1–6).
- 17. Create a Playlist class that stores song titles in a vector and prints them.
- 18. Create a Fraction class with numerator and denominator and a function simplify().
- 19. Create a BankAccount class that uses const correctly in getter functions.
- 20. Create a Shape base class and derive Rectangle and Circle from it (basic inheritance).